I. Assumed Parameters

a. Radiating/absorbing area of avionics enclosure: $A = 0.12\ \text{m}^2$ (external, top and partial side view)
b. Optical/thermal properties (uninsulated painted enclosure, conservative high-emissivity case): solar absorptivity $\alpha = 0.20$, emissivity $\varepsilon = 0.85$
c. Solar irradiance at Europa (~5.2 AU): $S_{Europa} \approx \frac{1361}{5.2^2} \approx 50\ \text{W/m}^2$
d. Albedo: Europa surface albedo ≈ 0.64 (one of the most reflective bodies in the Solar System), effective view factor ≈ 0.3; $S_{alb,eff} \approx 0.64 \times 0.3 \times 50 \approx 9.7\ \text{W/m}^2$
e. Planetary IR to the node (effective): Europa surface ≈ 90 K $\Rightarrow \sigma T^4 \approx 3.7\ \text{W/m}^2$; $S_{IR,eff} \approx 3.7\ \text{W/m}^2$ (essentially negligible)
f. Internal heat (hot/operating case, avionics + payload electronics): $Q_{int,hot} = 20$ W
g. Internal heat (cold/survival case at night, electronics only, pre-heater): $Q_{int,cold} = 5$ W
h. Stefan-Boltzmann constant: $\sigma = 5.67 \times 10^{-8}\ \text{W/m}^2\text{K}^4$

II. Methods

a. Gain:
   i. $Q_{sun} = \alpha S_{Europa} A$
   ii. $Q_{alb} = \alpha S_{alb,eff} A$
   iii. $Q_{IR} = \varepsilon S_{IR,eff} A$
   iv. $Q_{int} = (given)$
b. Loss: $Q_{out} = \varepsilon \sigma A T^4$
c. Equilibrium: $Q_{in} = Q_{sun} + Q_{alb} + Q_{IR} + Q_{int}$, $\quad T = \left(\frac{Q_{in}}{\varepsilon \sigma A}\right)^{1/4}$

III. Hot Scenario (daytime operations)

a. $Q_{sun} = \alpha S_{Europa} A = 0.20 \times 50 \times 0.12 = 1.21$ W
b. $Q_{alb} = \alpha S_{alb,eff} A = 0.20 \times 9.7 \times 0.12 = 0.23$ W
c. $Q_{IR} = \varepsilon S_{IR,eff} A = 0.85 \times 3.7 \times 0.12 = 0.38$ W
d. $Q_{int} = 20$ W
e. $Q_{in} = 1.21 + 0.23 + 0.38 + 20 = 21.82$ W
f. $T_{hot} = \left(\frac{21.82}{0.85 \times 5.67 \times 10^{-8} \times 0.12}\right)^{1/4} \approx 248\ \text{K} = -25\ ^\circ\text{C}$

IV. Cold Scenario (night survival)

a. $Q_{sun} = 0$
b. $Q_{alb} = 0$
c. $Q_{IR} = \varepsilon S_{IR,eff} A = 0.85 \times 3.7 \times 0.12 = 0.38$ W
d. $Q_{int} = 5$ W
e. $Q_{in} = 0.38 + 5 = 5.38$ W
f. $T_{cold} = \left(\frac{5.38}{0.85 \times 5.67 \times 10^{-8} \times 0.12}\right)^{1/4} \approx 175\ \text{K} = -98\ ^\circ\text{C}$

V. Reflection

This Europa surface point-mass estimate shows a ∼73 K swing between daytime operation (∼248 K, −25 °C) and night survival (∼175 K, −98 °C). Unlike a Mars-class mission, the problem on Europa is not rejecting heat but retaining it: solar irradiance at 5.2 AU collapses to ∼50 W/m^2, so internal dissipation dominates the energy balance and even the operating case falls below the ∼253 K (−20 °C) lower bound for Li-ion battery charging. The survival case sits far outside any electronics or battery limit and would freeze the system without intervention. This drives three coupled design decisions: MLI to reduce the effective emissivity of the enclosure and suppress radiative loss at night, routed RTG waste heat as the primary always-on thermal source (the MMRTG-class unit rejects on the order of 2 kW thermal continuously), and controlled survival heaters to hold the batteries within their operational band. Because heater energy competes directly with the 110 W continuous RTG electrical budget, thermal and power design must be co-optimized: better insulation lowers heater demand, which frees power for science and mobility. These results justify requirements for battery temperature limits, a bounded survival heater load, and thermal-vacuum verification at 90 K before flight.

AI was used to fact-check the results.